An-Najah National University



Wireless Networks Project Report

Physiological-Signal-Based Key Agreement for Wireless Body Area Networks (IEEE 802.15.6)

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Abstract

Wireless Body Area Networks (WBANs) operating under the IEEE 802.15.6 standard must balance strict energy efficiency with strong security. Traditional cryptographic key exchange protocols require significant wireless communication, which drains the batteries of implanted medical sensors and increases the risk of exposing sensitive health data. This project evaluates a physiological-signal-based key agreement protocol. Two sensors on the same body extract synchronized heartbeat features, specifically inter-pulse intervals (IPIs), from the ECG signal and generate the cryptographic key locally, so the key is never transmitted over the air. A Bloom filter reconciles the two sensors' feature sets without revealing the raw biometric values, and HMAC tags confirm that both sides hold the same key. Simulation results on the MIT-BIH Arrhythmia Database show that both sensors derive an identical 160-bit key with matching HMAC verification. The Python prototype completes in about 7.04 seconds, most of which is ECG processing. We also validated the Bloom filter experimentally, and found and fixed a defect that had raised its false-positive rate from the 0.096% target to 49.5%. The approach removes over-the-air key exchange while keeping the communication overhead to about 100 bytes, which makes it suitable for battery-constrained WBAN devices.

Keywords: WBAN, IEEE 802.15.6, ECG, inter-pulse interval, Bloom filter, key agreement, HMAC.

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1. Introduction

Wireless Body Area Networks (WBAN), defined by the IEEE 802.15.6 standard [1], are designed for medical applications where tiny sensors are deployed on or inside the human body [2]. Because these medical devices have strictly limited battery capacities and transmit highly sensitive health data, maintaining robust security while ensuring maximum energy efficiency are critical and often conflicting challenges.

IEEE 802.15.6 offers three security levels: unsecured communication, authentication only, and authentication with encryption (AES-128 in CCM mode). Both secured levels need a shared master key between the node and the hub, which the standard establishes through elliptic-curve Diffie–Hellman based association. These procedures cost extra message exchanges and public-key computation on the sensor.

Every sensor on the same body observes the same heartbeat. The time between consecutive beats, the Inter-Pulse Interval (IPI), is random enough to serve as key material, hard to measure from a distance, and identical everywhere on the body [3]. Schemes such as PSKA [4] have shown that physiological signals let co-located sensors agree on a key without any pre-deployed secret.

1.1 Problem Statement

The traditional cryptographic key exchange protocols (such as Diffie–Hellman [5]) require significant wireless communication overhead. In WBANs, wireless transmission is usually the most energy-consuming operation. Exchanging keys over the air rapidly drains the medical sensors' batteries, which can lead to required surgical replacements for implanted devices, and exposes the highly sensitive medical data to interception and Man-in-the-Middle (MitM) attacks.

The goal is to let two sensors on the same body, S (initiator) and R (responder), establish a shared key such that:

G1. the key is never transmitted over the wireless channel;

G2. the raw biometric data (IPI values) is never transmitted in the clear;

G3. both sensors can confirm that they hold the same key;

G4. communication and computation stay low enough for constrained sensors.

1.2 Threat Model

We assume an adversary who can eavesdrop on, replay, drop and modify every wireless message between S and R, but who is not in physical contact with the patient and

therefore cannot measure the patient's ECG at the same moment as the sensors. A compromised sensor is outside the scope of this project.

2. System Description

To resolve the energy–security trade-off in WBANs, this project implements and evaluates a Physiological-Signal-Based Key Agreement protocol, building on the open-source `physio-key-gen` implementation [6]. The system model consists of a sender (Sensor S) and a receiver (Sensor R) located on the same patient's body.

Rather than transmitting cryptographic keys over the air, both sensors independently generate the same key locally from synchronous heartbeat features, the Inter-Pulse Intervals (IPI), extracted from the patient's ECG signal. Because both sensors observe the same physiological process, their IPI readings are strongly correlated. A Bloom filter [7] reconciles small reading discrepancies without exposing the raw biometric data, so the key stays on the sensors. The protocol has three phases, summarised in Figure 1.

2.1 Phase 1: Feature Generation

- **QRS detection.** A 3-second window of the ECG (lead MLII, $f_s = 360$ Hz) is cross-correlated with a 5-sample sinusoidal template. Samples whose correlation exceeds a threshold $\theta = 2$ are marked as R-peaks.
- **IPI.** The IPI is the average time between consecutive detected peaks: IPI $=$ mean$\{(p_{j+1} - p_j)/f_s\}$.
- **Quantisation.** The IPI is converted to a fixed-point binary string with 13 fractional bits. For example, IPI ≈ 0.882 s becomes `.1110000111100`.
- **Feature vector.** Each feature f_i concatenates three quantised IPIs from windows shifted by one sample. Each sensor builds $N_f = 30$ features: F_S at S and F_R at R.

2.2 Phase 2: Reconciliation with a Bloom Filter

S inserts its 30 features into a Bloom filter sized for a false-positive rate p:

$$m = -\frac{n \ln p}{(\ln 2)^2}, \qquad k = \frac{m}{n} \ln 2. \tag{1}$$

With $n = 30$ and $p = 1/1040$ this gives $m = 433$ bits and $k = 10$ SHA-1 based hash functions. S sends $\langle ID_S, BF_S, N_o \rangle$, where N_o is a random nonce. R tests each of its own features against the filter, and the matching positions form the common index set I. The filter contains only hash positions, so an eavesdropper learns neither F_S nor F_R.

2.3 Phase 3: Key Generation and Confirmation

1. If $|I|$ exceeds the security threshold $\tau = 8$, R computes $K_R = \text{SHA1}(\|_{i \in I} f_i^R)$ and $MAC_1 = \text{HMAC}_{K_R}(I \,\|\, N_o)$ [8], then sends $\langle ID_R, MAC_1, I \rangle$.
2. S computes $K_S = \text{SHA1}(\|_{i \in I} f_i^S)$ and $MAC_2 = \text{HMAC}_{K_S}(I \,\|\, N_o)$. If $MAC_1 = MAC_2$, then $K_S = K_R$, I was not modified, and the reply is fresh.
3. S sends the confirmation $C = \text{HMAC}_{K_S}(ID_S \,\|\, ID_R \,\|\, N_o)$. R verifies it with K_R, and both sides accept the key.

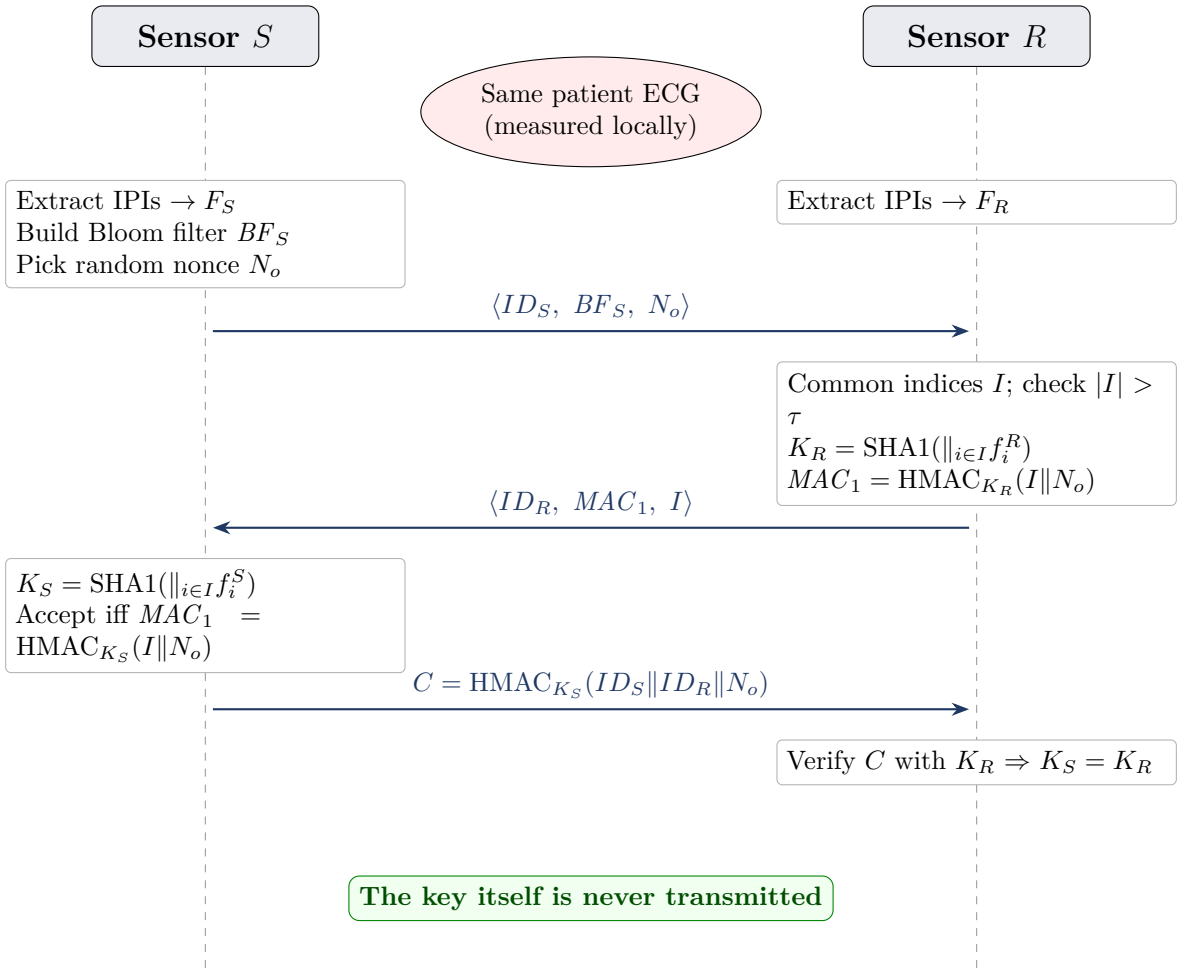


Figure 1: Message flow of the key agreement protocol between sensors S and R.

3. Cross-Layer Optimization for Energy Efficiency

Within the context of the IEEE 802.15.6 standard, optimizing the interaction between the security sub-layer and the Physical/MAC layers is essential for prolonging network lifetime.

Our protocol takes a cross-layer approach: it shifts the energy burden of key establishment from the radio transceiver to the local microcontroller. The key material comes from a physical-world signal (the patient's ECG) that both sensors already measure for their medical function, instead of from public-key messages exchanged over the MAC layer. The resulting 160-bit shared secret can serve as the master key from which the IEEE 802.15.6 security sub-layer derives the AES-128 keys it uses to protect frames.

This design brings three benefits:

- **Shorter radio activity.** Key establishment needs only three short messages, about 100 bytes of payload in total (Section 5). No public keys or certificates have to be sent.
- **No public-key computation.** The sensors use only SHA-1 and HMAC, which are far cheaper than the elliptic-curve operations of the standard's association procedures.
- **No pre-deployment.** No password or certificate has to be loaded into the sensor before use, because the shared secret comes from the body itself.

Together these reduce the radio duty cycle needed for key establishment, which helps extend the battery life of energy-constrained medical implants.

4. Performance Analysis

The simulation was implemented in a Python environment based on the open-source `physio-key-gen` repository [6]. The implementation uses the `wfdb` library to process real medical ECG data from the MIT-BIH Arrhythmia Database (Record 103) [9], distributed through PhysioNet [10]. Keys are derived with SHA-1 and verified with HMAC, and an execution time tracker measures the computational cost of the algorithm. Table 1 lists the parameters used.

Table 1: Simulation parameters.

Parameter	Value
ECG record / lead	MIT-BIH 103 / MLII, 360 Hz
Analysis window	3 s (1080 samples)
Peak-detection threshold	2
Fractional bits per IPI	13
Number of features N_f	30
Bloom filter	$m = 433$ bits, $k = 10$, target $p = 1/1040$
Security threshold τ	8 common features
Hash / MAC	SHA-1 / HMAC-SHA1 (160-bit)
Software	Python 3, `wfdb`, `numpy`, `bitarray`, `hashlib`, `hmac`

4.1 Feature Extraction

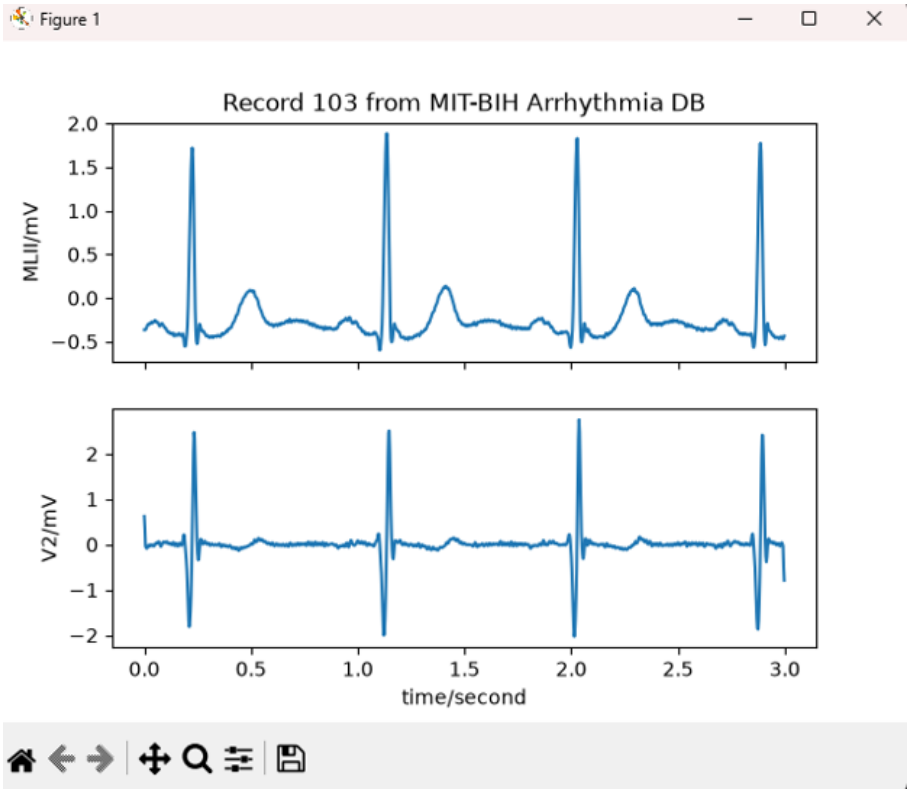


Figure 2: Extraction of physiological features (Inter-Pulse Intervals) from the MIT-BIH Arrhythmia Database, Record 103.

Figure 2 shows the ECG signal processed by the sensors. The QRS complexes are clear and regular, so R-peak detection is reliable. The extracted IPI is about 0.882 s (binary `.1110000111100`), which corresponds to a heart rate of about 68 beats per minute. Figure 3 shows the 30 feature vectors generated independently by S and R.

4.2 Key Agreement

As shown in Figure 4 and Table 2, all 30 features of R passed the Bloom filter test ($|I| = 30 > \tau$). Both sensors generated the identical key and matching HMAC tags without any key transmission, and the final confirmation succeeded.

4.3 Bloom Filter Validation

To check that the Bloom filter meets its design target, we inserted 30 random 42-bit features into a filter with the parameters in Table 1. We then queried it with 200 000 random features that had not been inserted. The original membership test in the repository returned after checking only the first of the $k = 10$ hash positions. Its false-positive rate was 49.5%, which would let guessed features pass reconciliation half of the time. After we

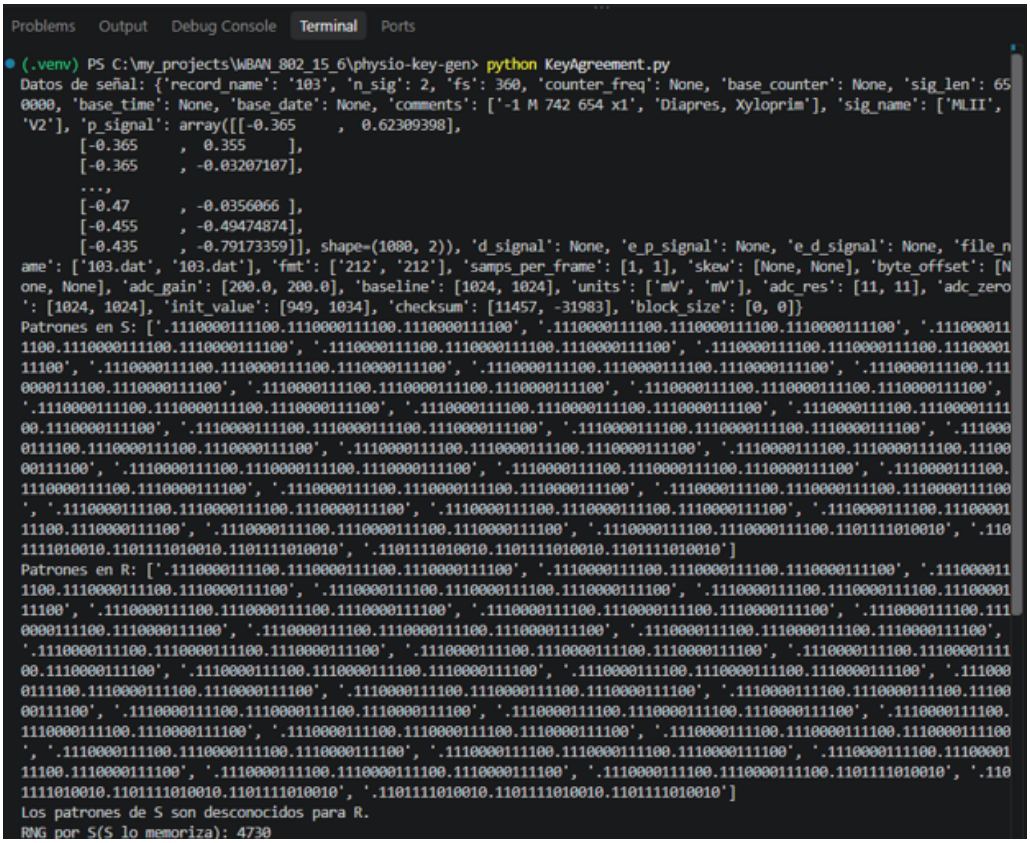
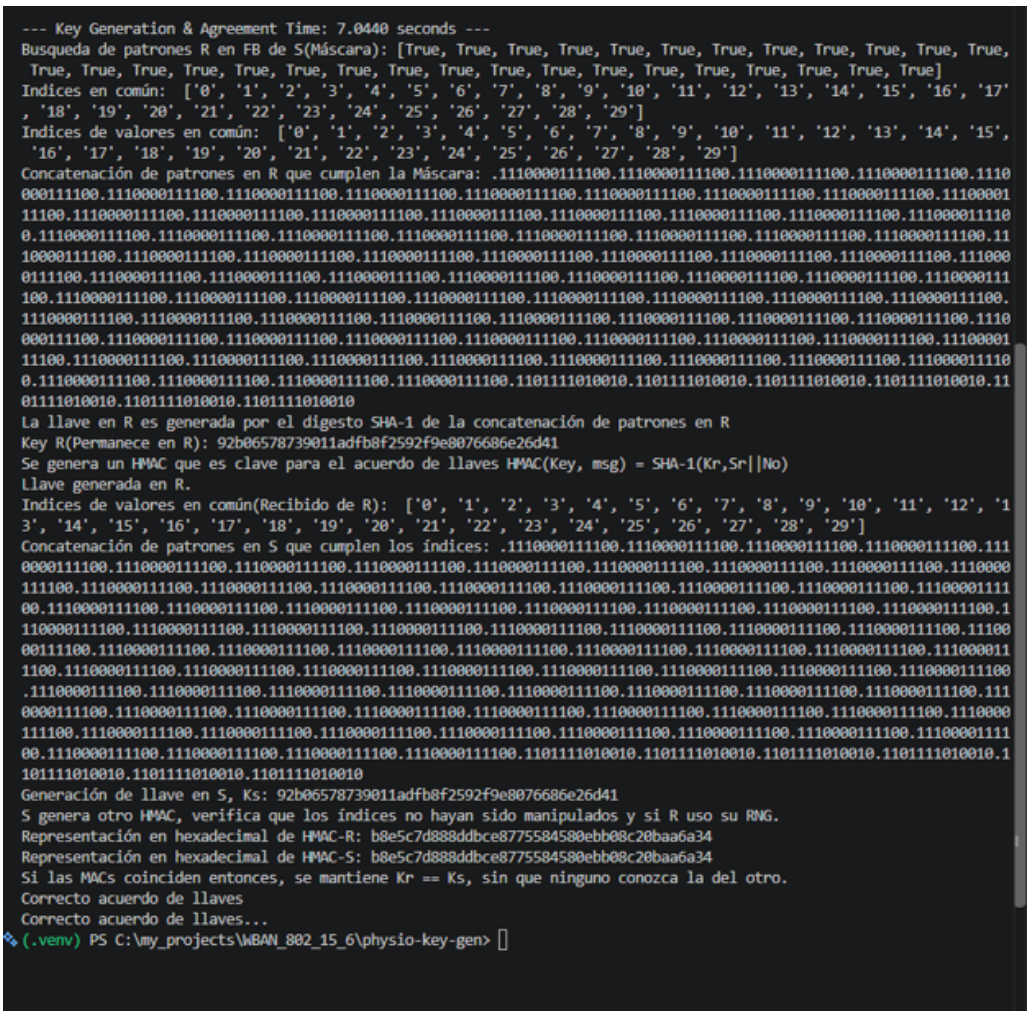


Figure 3: Terminal output: ECG record metadata and the feature vectors generated by sensors S and R.

Table 2: Key agreement results (hexadecimal, 160-bit).

Value	**Result**
Key at R, K_R	`92b06578739011adfb8f2592f9e8076686e26d41`
Key at S, K_S	`92b06578739011adfb8f2592f9e8076686e26d41`
MAC_1 (from R)	`b8e5c7d888ddbce8775584580ebb08c20baa6a34`
MAC_2 (at S)	`b8e5c7d888ddbce8775584580ebb08c20baa6a34`
Nonce N_o	4730
Final confirmation	Verified


--- Key Generation & Agreement Time: 7.0440 seconds ---
Busqueda de patrones R en FB de S(Máscara): [True, True]
Indices en común: ['0', '1', '2', '3', '4', '5', '6', '7', '8', '9', '10', '11', '12', '13', '14', '15', '16', '17', '18', '19', '20', '21', '22', '23', '24', '25', '26', '27', '28', '29']
Indices de valores en común: ['0', '1', '2', '3', '4', '5', '6', '7', '8', '9', '10', '11', '12', '13', '14', '15', '16', '17', '18', '19', '20', '21', '22', '23', '24', '25', '26', '27', '28', '29']
Concatenación de patrones en R que cumplen la Máscara: .1110000111100.1110000111100.1110000111100.

corrected it to check all k positions, the measured rate dropped to 0.093%, matching the design target (Table 3). The protocol still produced the same key after the fix.

Table 3: Measured Bloom filter false-positive rate ($m = 433$, $k = 10$, $n = 30$, 200 000 probes).

Membership test	False negatives	False-positive rate
Design target (Eq. 1)	0	0.096%
Original: checks only the first hash	0	49.5%
Corrected: checks all k hashes	0	**0.093%**

4.4 Execution Time

The execution time tracker recorded 7.04 seconds for the key generation stage of the Python prototype. This covers loading the ECG record, extracting the IPI features for both sensors, and building the Bloom filter on a desktop computer. Most of the time is spent on file input/output and signal processing in interpreted Python. The cryptographic operations themselves (a few SHA-1 and HMAC computations) take only microseconds. A native implementation on a sensor microcontroller would be considerably faster.

5. Security Analysis

Table 4 maps the goals from the problem statement to the mechanisms that achieve them.

Table 4: Security goals and how the protocol meets them.

Goal / attack	Mechanism
Key never on air (G1)	K_S and K_R are computed locally. Only the Bloom filter, the indices, the nonce and the MAC tags are transmitted.
Biometric privacy (G2)	The Bloom filter stores only hash positions. I reveals which features match, not their values.
Key confirmation (G3)	Matching MAC_1, MAC_2 and C prove that both sides hold the same key without revealing it.
Eavesdropping	Without the ECG features, an attacker cannot recompute the key from the observed messages.
Tampering with I	I is covered by MAC_1. Any change makes $MAC_1 \neq MAC_2$, and S aborts.
Replay	The fresh nonce N_o is bound into every MAC, so tags from old sessions fail.
Man-in-the-Middle	An attacker away from the body cannot measure the same ECG, so it cannot produce a valid MAC_1 or C.

Table 5: Approximate payload of each protocol message (excluding device IDs).

Message	Content	Payload
$S \to R$	Bloom filter (433 bits) + nonce	$\approx$ 58 B
$R \to S$	MAC_1 (160 bits) + index bitmap (30 bits)	$\approx$ 24 B
$S \to R$	Confirmation tag C (160 bits)	20 B
Total		$\approx$ 102 B

5.1 Communication Overhead

As Table 5 shows, the whole agreement needs about 100 bytes of payload in three messages, and the key itself is never among them.

5.2 Limitations

- **Same-signal simulation.** Both sensors read the same ECG lead from the same start sample, so their features are identical. Real sensors would see noise and time offsets, and future tests should measure false-accept and false-reject rates under those conditions.
- **Feature entropy.** Neighbouring windows are shifted by only one sample, so many features repeat. The effective entropy of the key is therefore lower than its 160-bit length. Longer, non-overlapping windows would improve it [3].
- **SHA-1.** SHA-1 is no longer collision-resistant [11]. HMAC-SHA1 is still considered acceptable, but key derivation should move to SHA-256.
- **Energy.** Energy savings are argued analytically. They should be measured on real sensor hardware.

6. Conclusion

This project demonstrates that physiological signals can be used to generate secure, synchronized cryptographic keys in WBANs. By using local biometric data (ECG inter-pulse intervals) and a Bloom filter for reconciliation, sensors S and R derived an identical 160-bit key and confirmed it with matching HMAC tags, without the key ever being transmitted. The protocol fits the power constraints of IEEE 802.15.6: it replaces over-the-air key exchange with about 100 bytes of messages and inexpensive hash operations, which defeats passive eavesdropping and, through nonce-bound HMACs, prevents tampering, replay and MitM attacks from an attacker away from the body. Our validation also found and fixed a Bloom filter defect, bringing its false-positive rate from 49.5% down to the

0.096% design target. Future work includes testing with separate, noisy sensors, increasing feature entropy, moving to SHA-256, and measuring energy on real hardware.

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